

Physiology: Lung Mechanics – Pressures Acting on the System

PPOM 2 - Respiratory System

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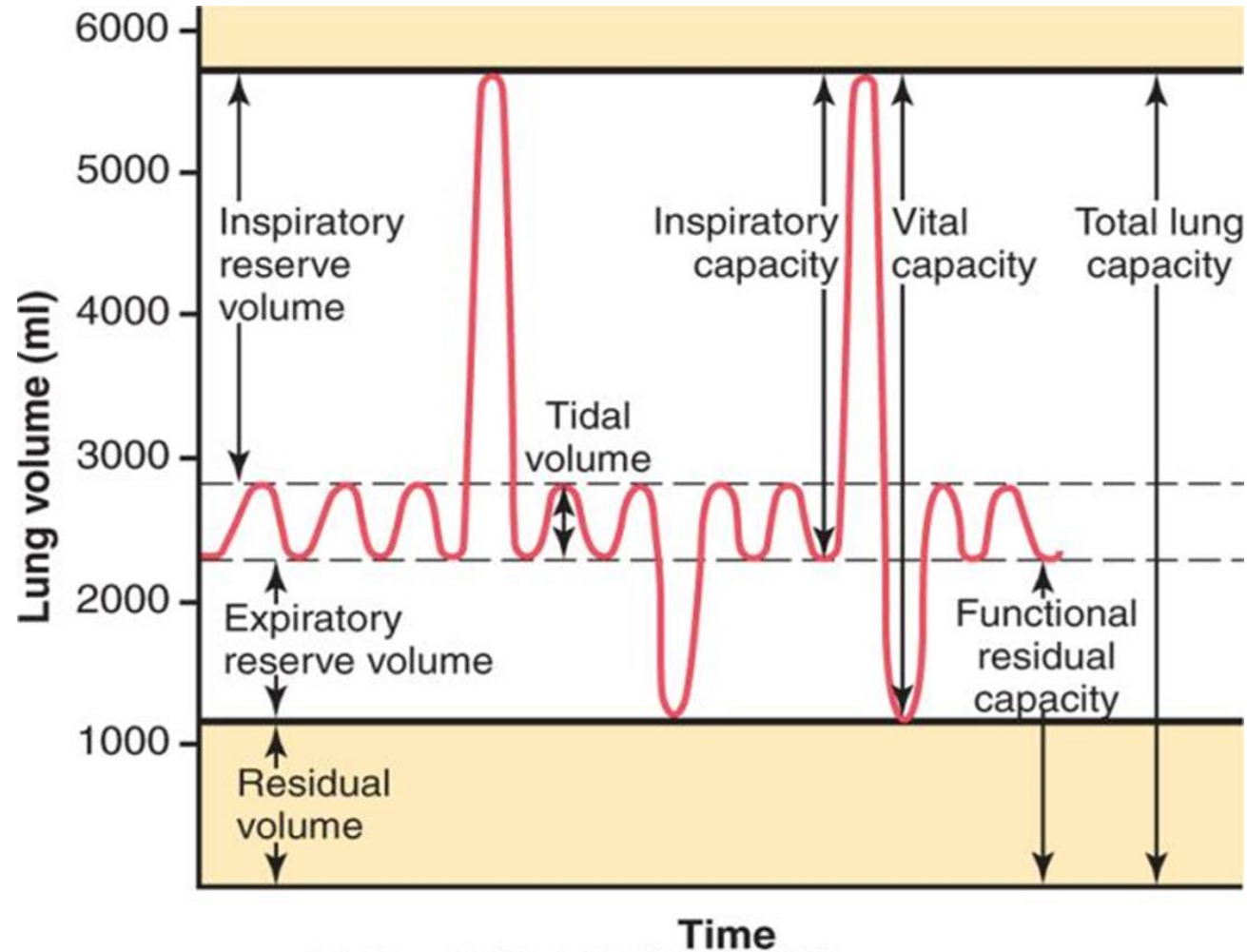
Learning Objectives

1. Distinguish between pulmonary & alveolar ventilation in normal respiratory physiology.
2. Apply muscle anatomy involved in the generation of pulmonary pressures.
3. Deconstruct lung mechanics generating intrapleural, alveolar and transpulmonary pressures.
4. Chart the functional relationship between intrapleural, alveolar and transpulmonary pressures.
5. Interpret relevant lung biomechanics in relation to restrictive and obstructive lung diseases.

Four Processes

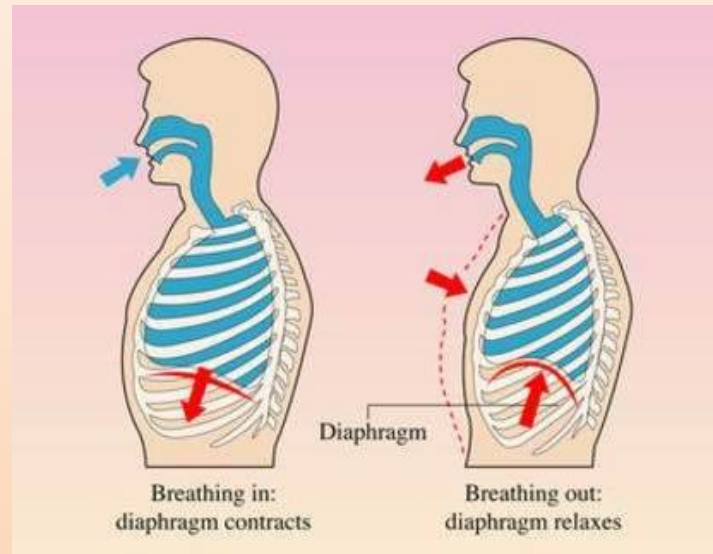
- Ventilation
- Diffusion
- Transport
- Regulation

Lung Volumes and Capacities



Ventilation

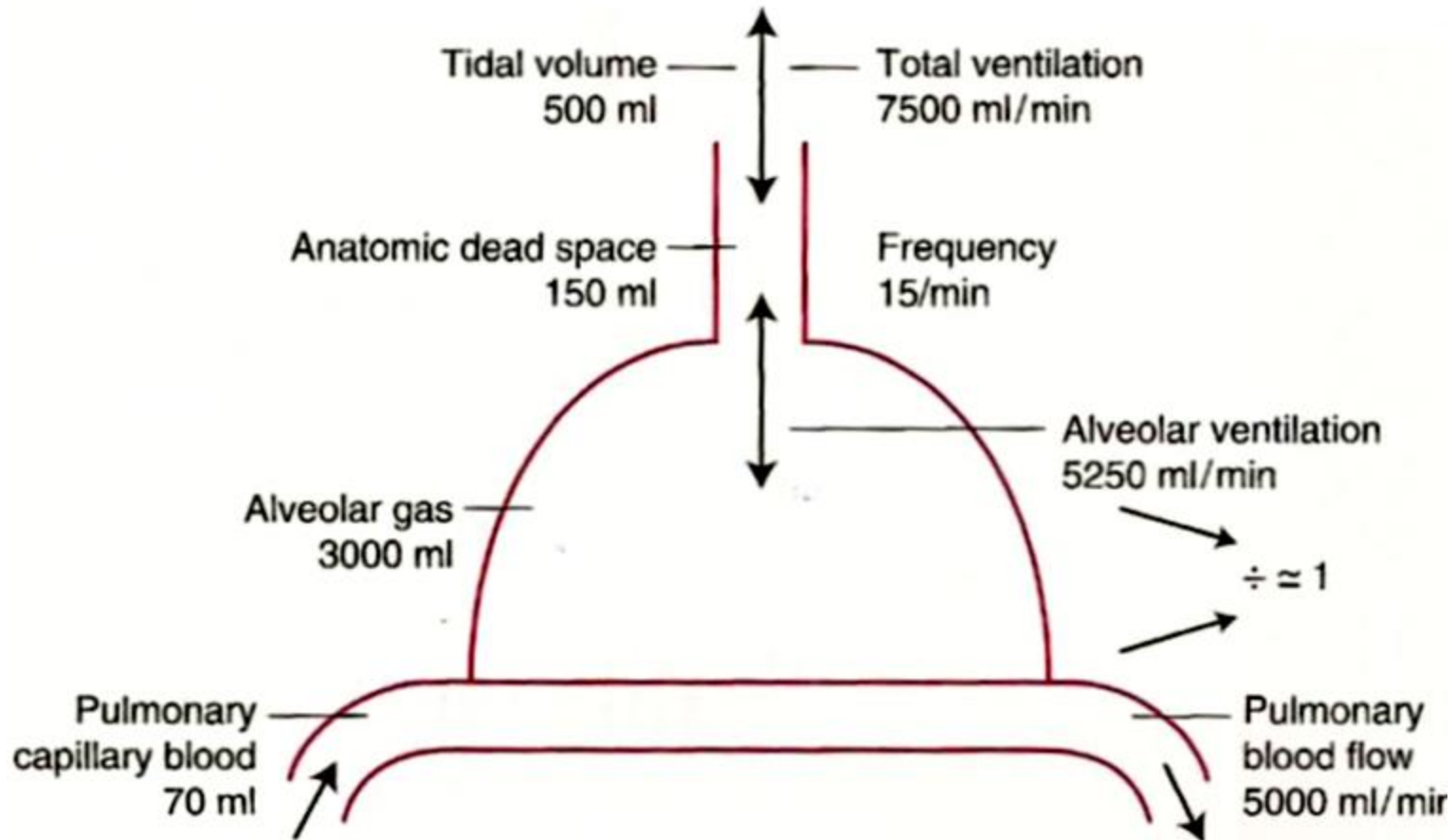
Ventilation occurs as a result of mechanical forces associated with the chest wall and lungs



Pulmonary and Alveolar Ventilation

- **Pulmonary Ventilation**
 - Gross ventilation per minute
 - **Tidal volume (V_T) x Respiratory Rate (RR)**
- **Alveolar Ventilation**
 - Fresh air reaching gas exchange areas of the lungs
 - **$(V_T - V_D) \times RR$**

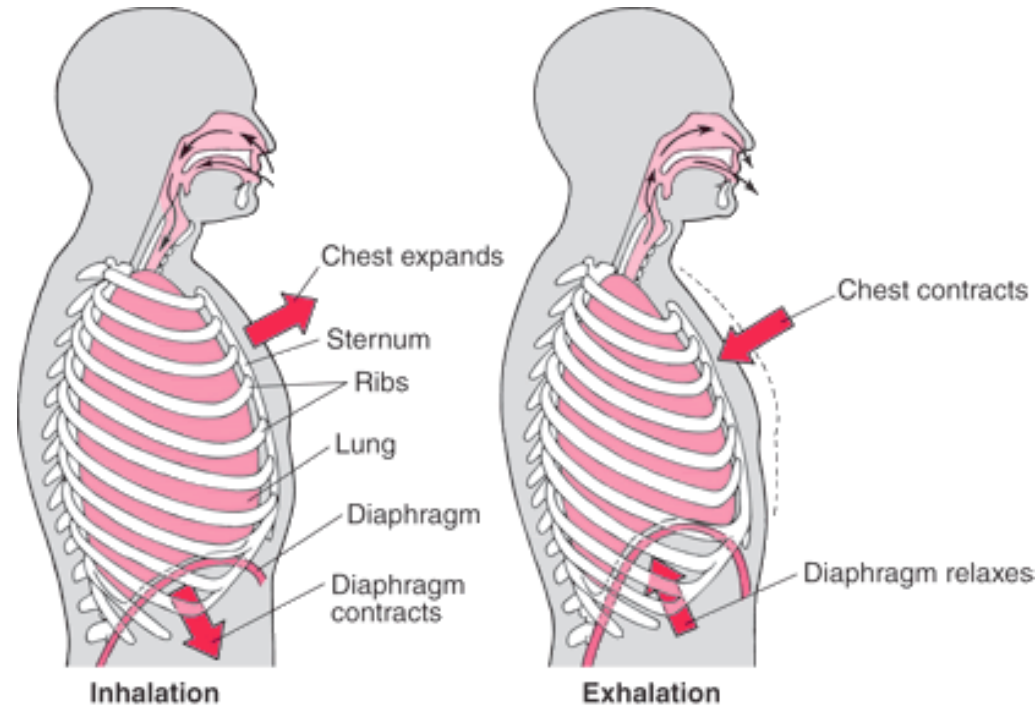
Summary - Ventilation



Pressures acting on the System

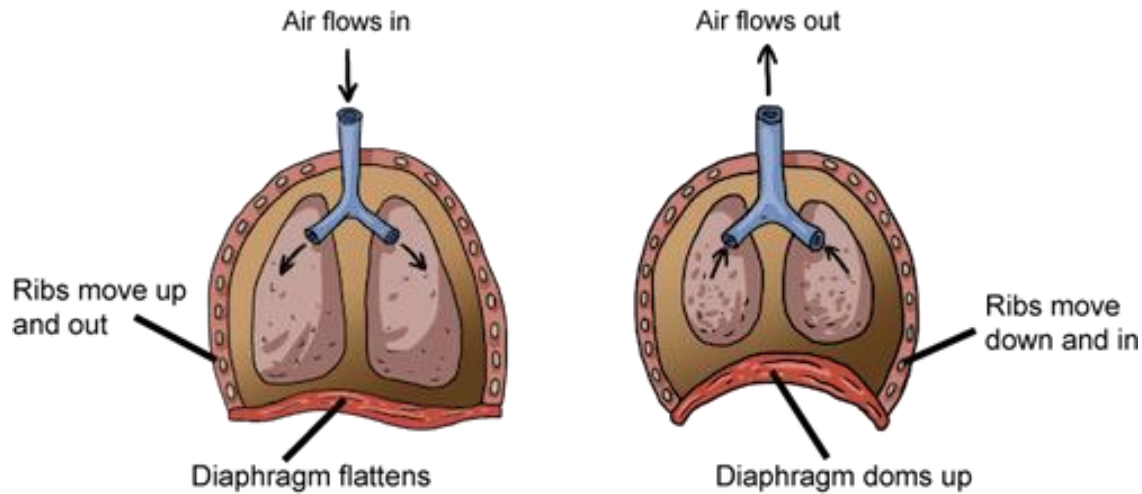
What Happens when You Breathe?

- Changes in volume of the rib cage and abdomen
- Change in pressure of the lungs

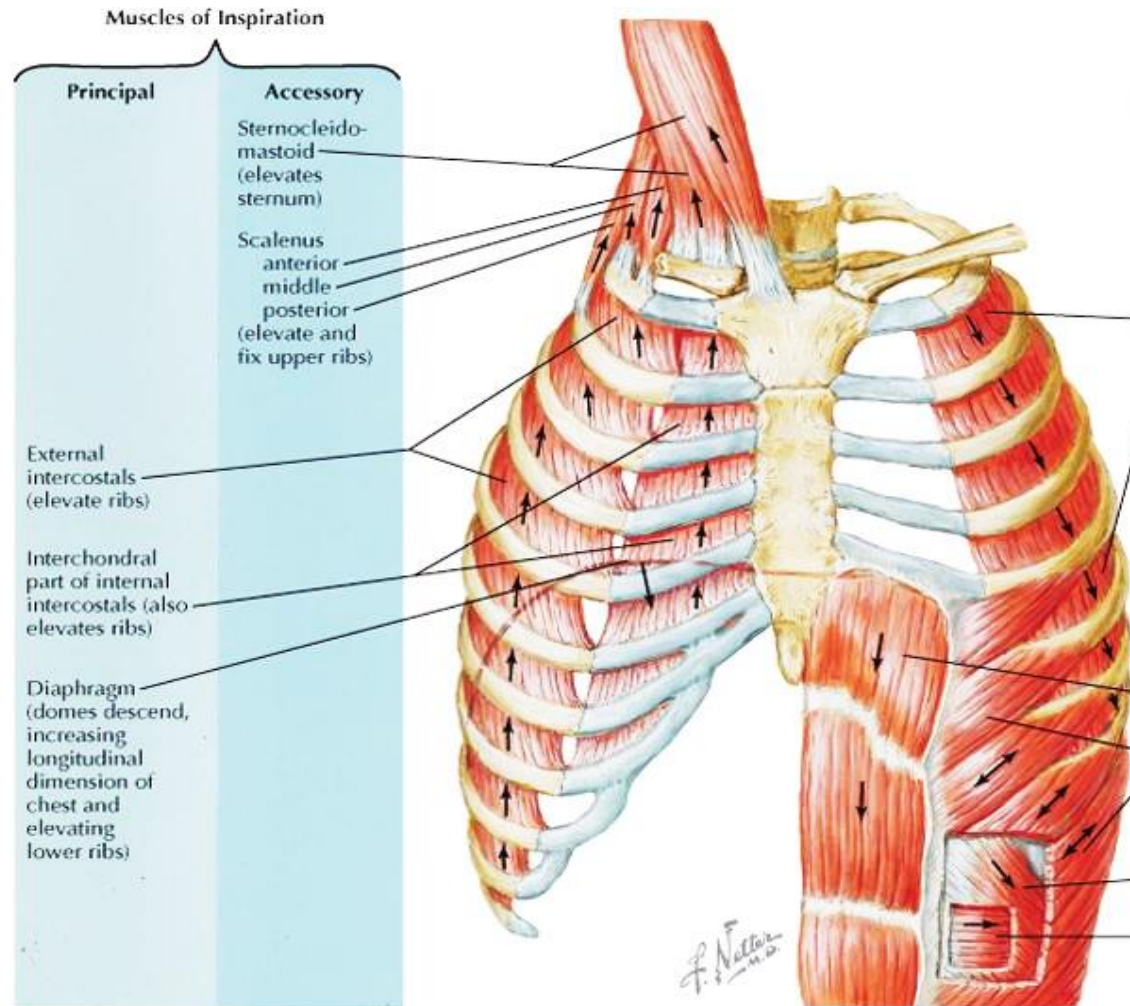


Muscles causing Lung Expansion and Contraction

Diaphragm: raising/lowering rib cage



Muscles of Inspiration



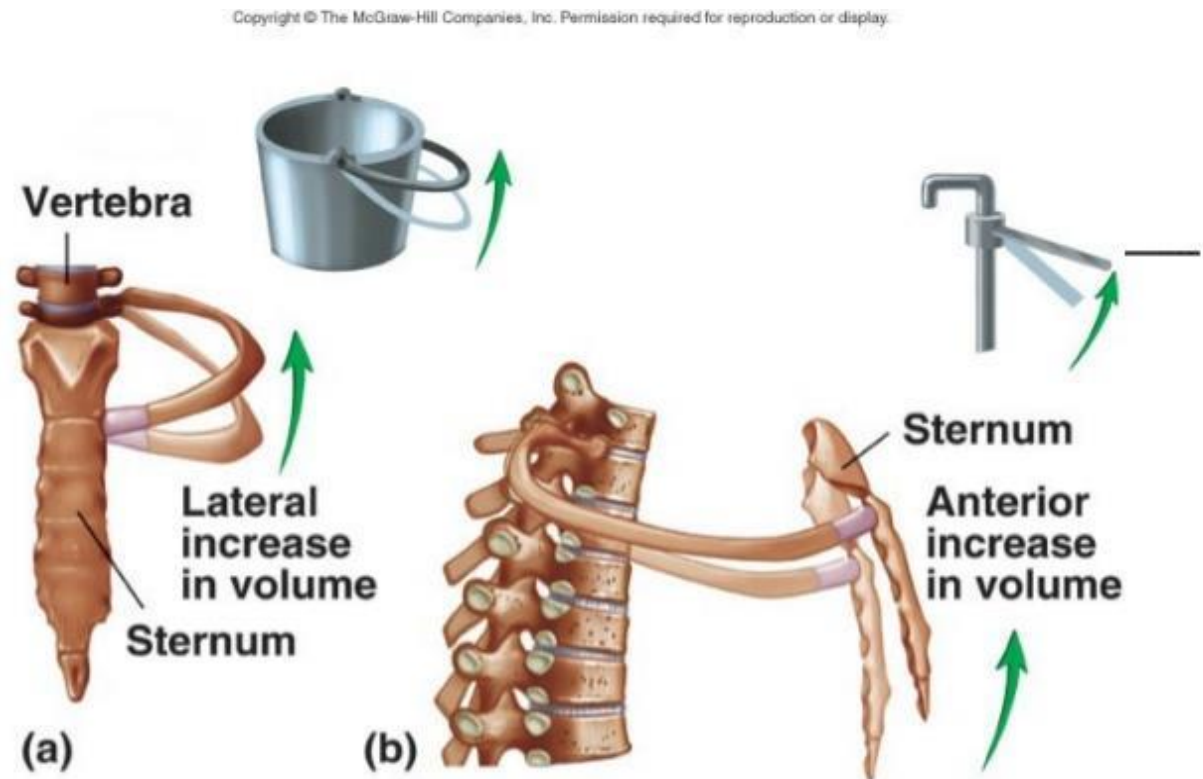
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A more detailed understanding here: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3297359/>

Muscles causing Lung Expansion and Contraction

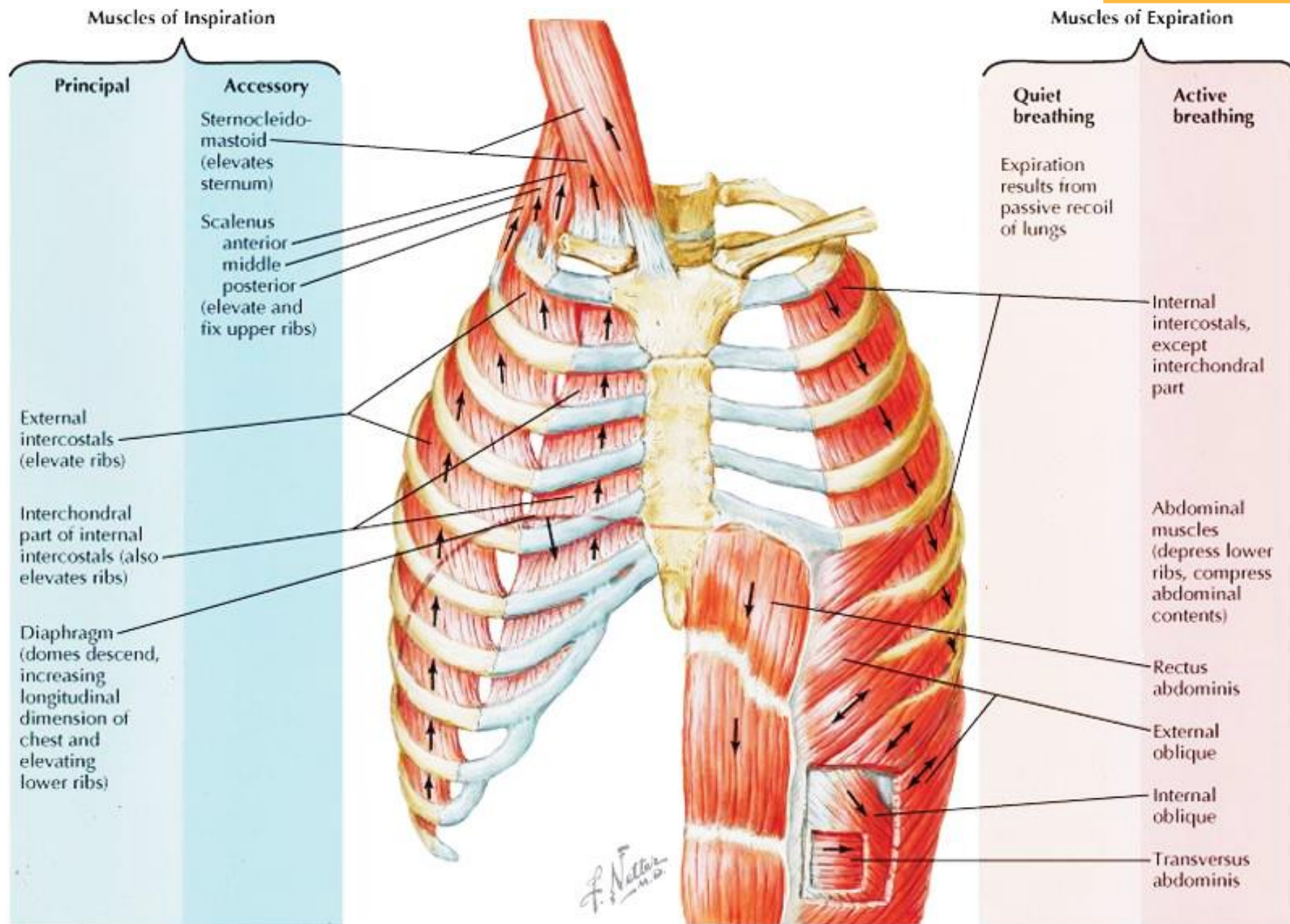
External Intercostals:

- Upward/forward movement of rib cage
- BUCKET HANDLE MOTION



- In **resting breathing**, inspiration is active, but expiration is passive (or mainly passive).
 - Diaphragm
- In **vigorous breathing**, both inspiration and expiration are active
 - **Inspiration:** Diaphragm + accessory muscles of inspiration
 - **Expiration:** Accessory muscles of expiration





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THE MUSCLES OF INSPIRATION AND EXPIRATION

The major muscles of inspiration and expiration are the following:

MUSCLES OF INSPIRATION

MINUTE VOLUME ⁽¹⁾

Diaphragm.....	Resting.....
External intercostals.....	> Resting.....
Sternocleidomastoids	
Anterior serrati plus scapular elevators	> 50 L/min
Scaleni.....	
Spinal erectus, trapezius, & posterior neck muscles.....	> 100 L/min

MUSCLES OF EXPIRATION

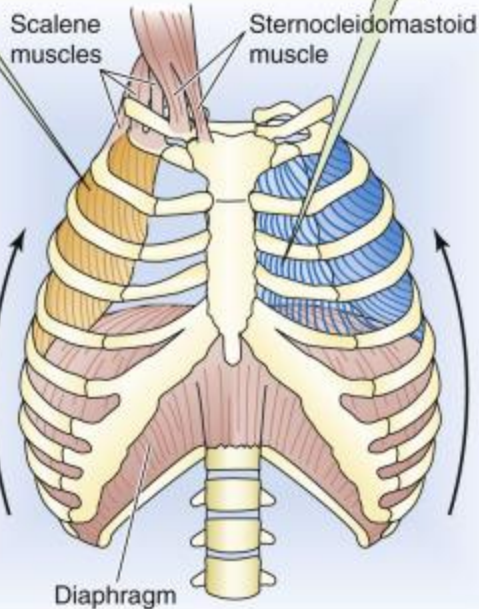
Internal intercostals.....	> Resting.....
Abdominals (recti, transversus, ext/int obliques)	} > 40 L/min (> 10 cm H ₂ O resist)
Posterior inferior serrati	

- (1) *Minute volume* is the amount of air breathed in and out (round trip) in one minute. The L/min numbers here indicate the minute volumes at which the various muscles begin to contribute to breathing in a healthy adult. In diseases where the airways are partially obstructed or movement of the lungs is restricted, the muscles become active at lower minute volumes.

Summary

A INSPIRATION

The most rostral and dorsal subsets of the **external** intercostal muscles (gold)—as well as the parasternal subset of the **internal** intercostal muscles (blue)—have an *inspiratory* mechanical advantage.

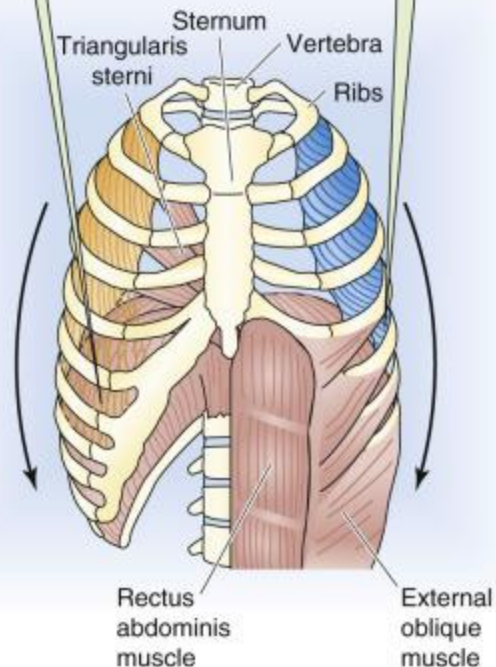


B BUCKET-HANDLE AND WATER PUMP-HANDLE EFFECTS



C EXPIRATION

The most caudal subset of the **internal** intercostal muscles (blue)—as well as the caudal-ventral subset of the **external** intercostal muscles (gold) and the triangularis sterni muscle (transversus thoracis)—have an *expiratory* mechanical advantage.



Factors affecting Diaphragm function

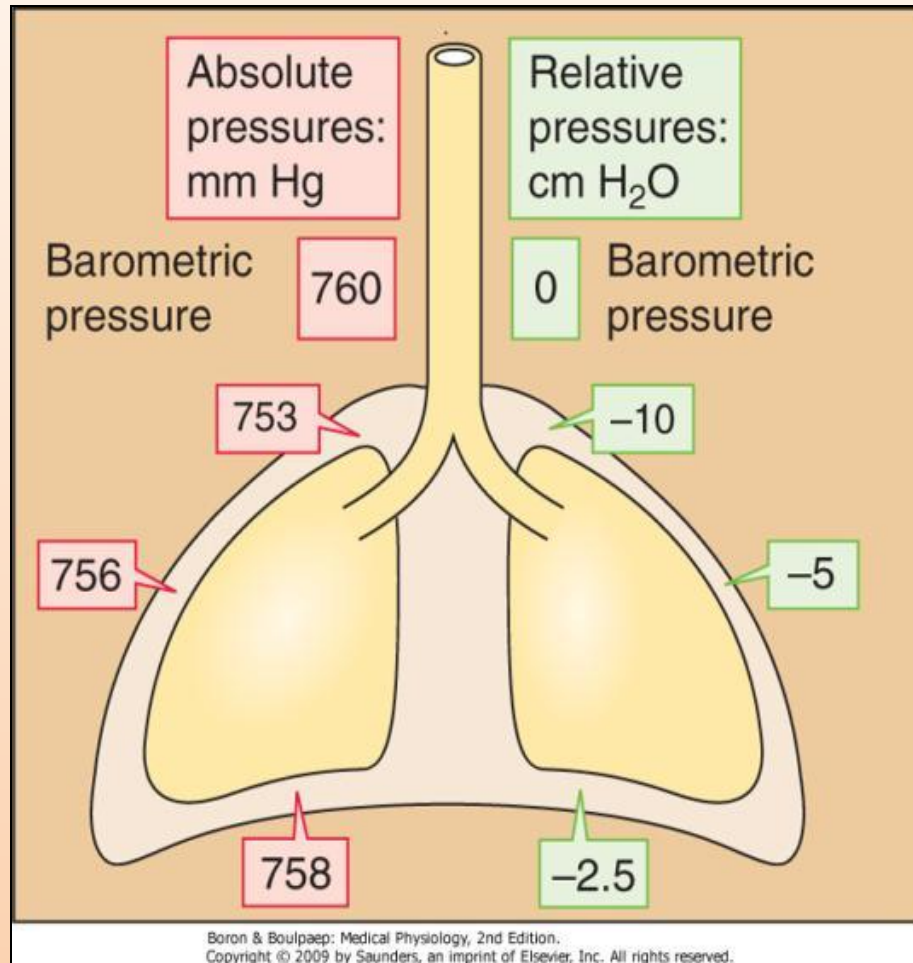
Paradoxical movement

Ascites

Pregnancy

Obesity

Pressures acting on the System

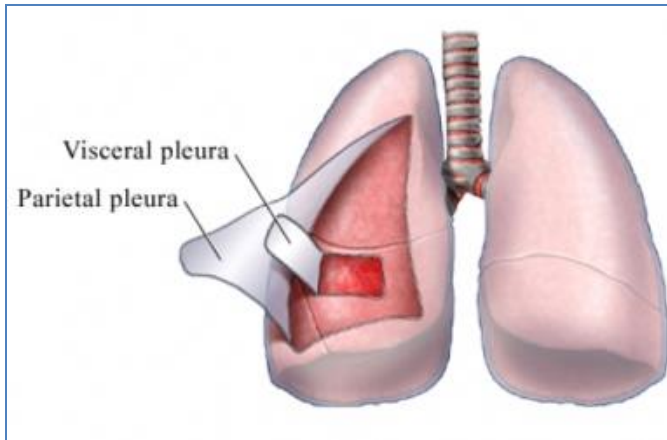


1. Intrapleural Pressure

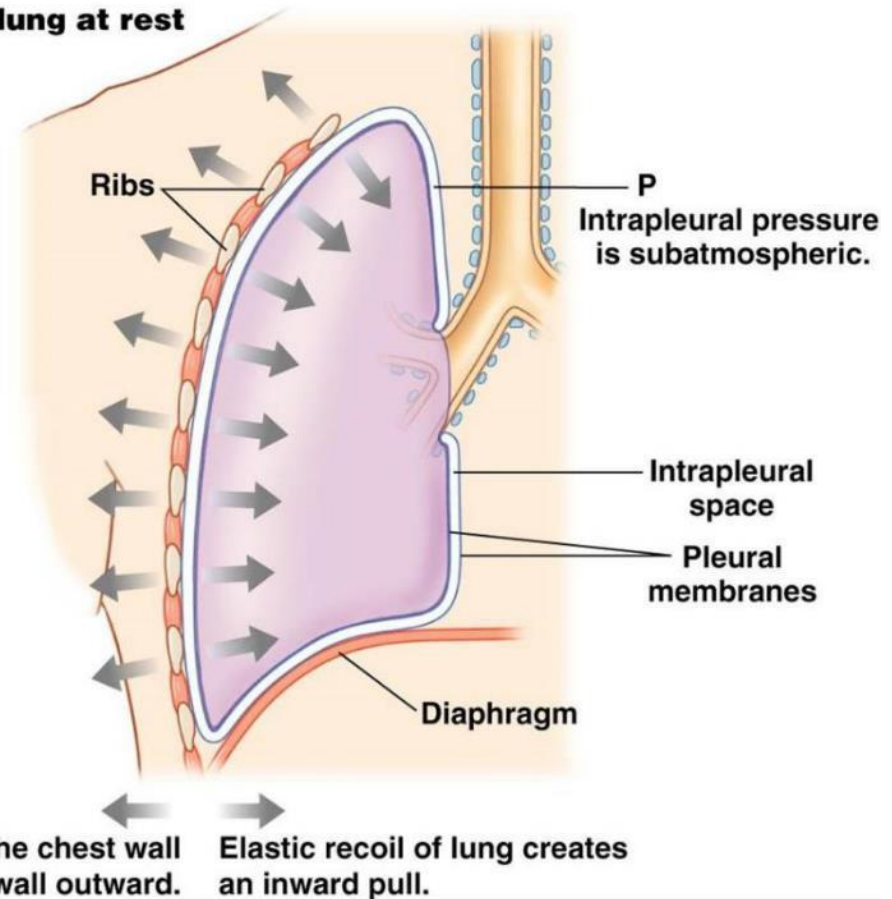
2. Alveolar Pressure

3. Transpulmonary Pressure

1. Intrapleural Pressure

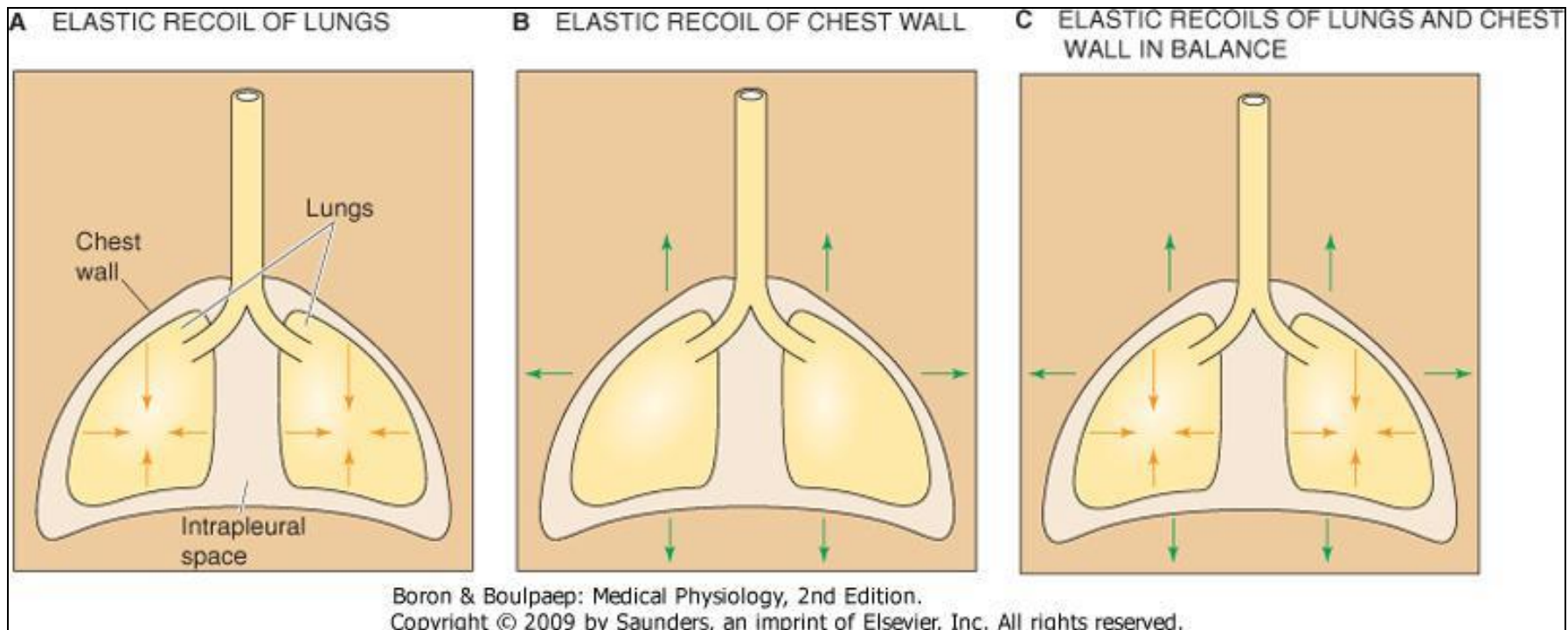


(a) Normal lung at rest

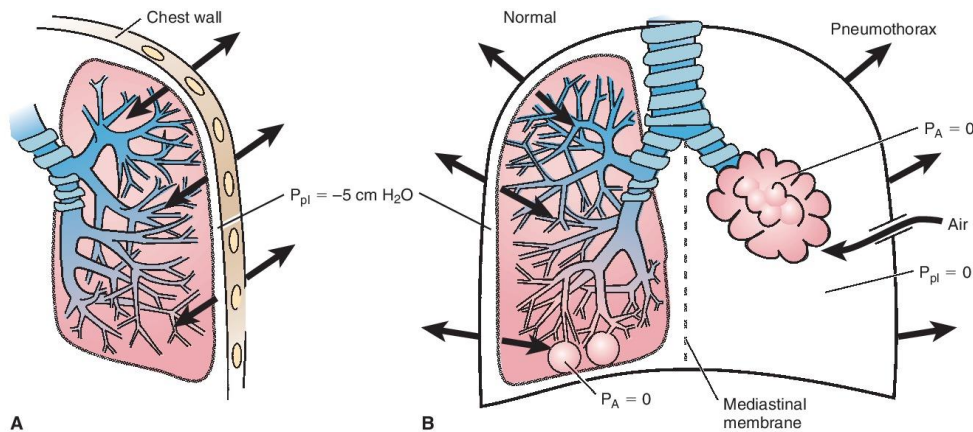


Negative Intrapleural Pressure

1. Lymphatic system drains the pleural fluid, generating a negative pressure
2. Elasticity of lungs and chest wall

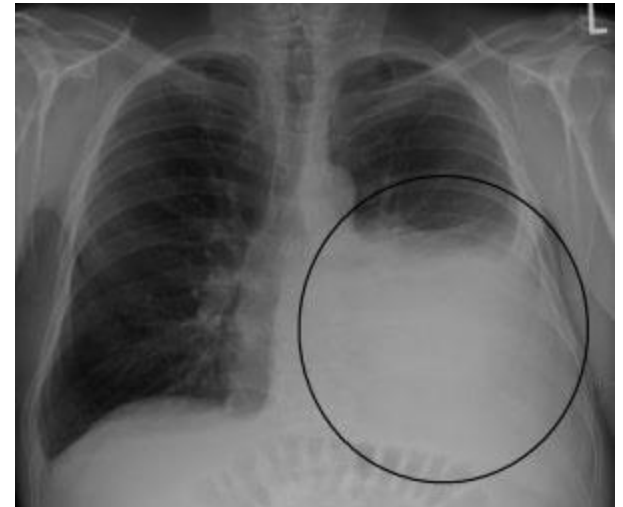


Negative Intrapleural Pressure

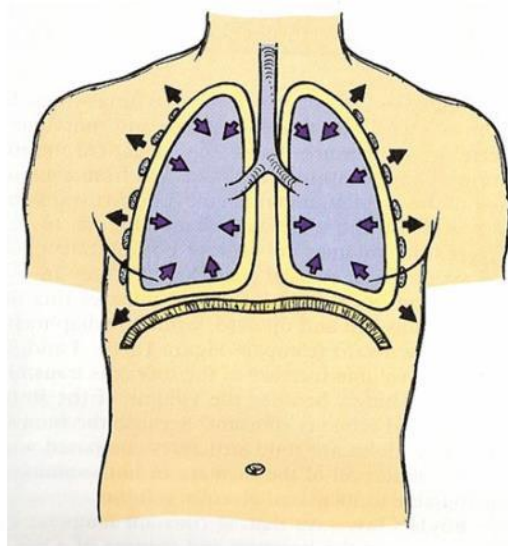


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The elastic recoil of the lungs and chest wall pulling in opposite directions create a negative pleural pressure: (A) The stretched lung (at the end of a normal inspiration) tends to recoil inwardly, and the chest wall tends to recoil outwardly, but in equal and opposite directions. Consequently, pleural pressure (P_{pl}) becomes negative (i.e., less than atmospheric pressure). (B) Rupture or puncture of the lung or chest wall results in a pneumothorax, during which the transpulmonary pressure becomes zero and elastic recoil causes the lung to collapse. The mediastinal membrane prevents the other lung from collapsing. P_a , alveolar pressure.

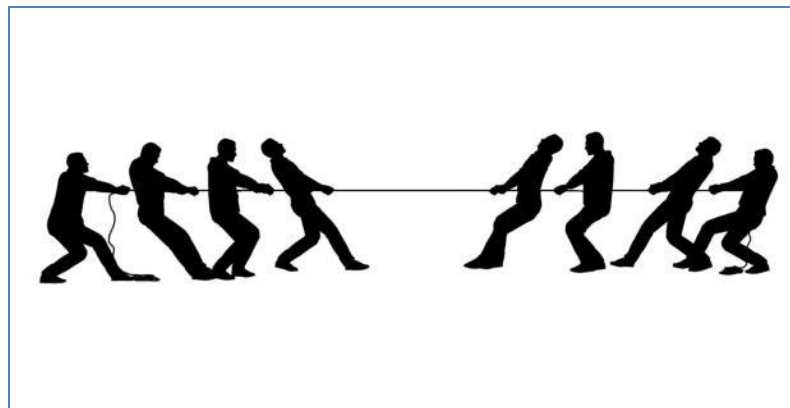


Both chest wall and lung have elastic recoil properties



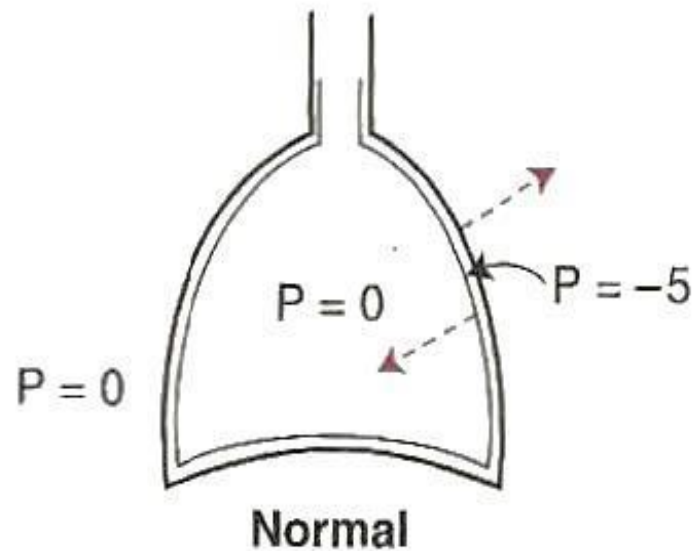
Outward elastic recoil pressure of the chest wall

Inward elastic recoil pressure of the lungs



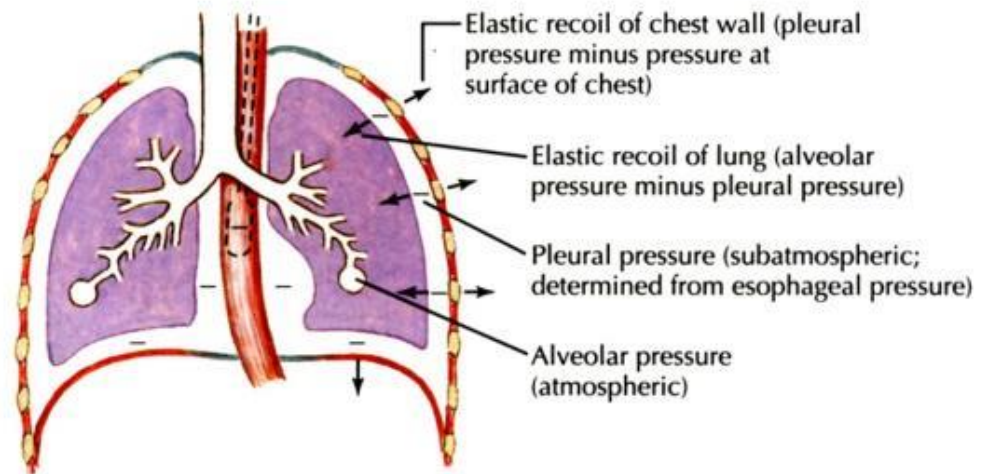
2. Alveolar Pressure

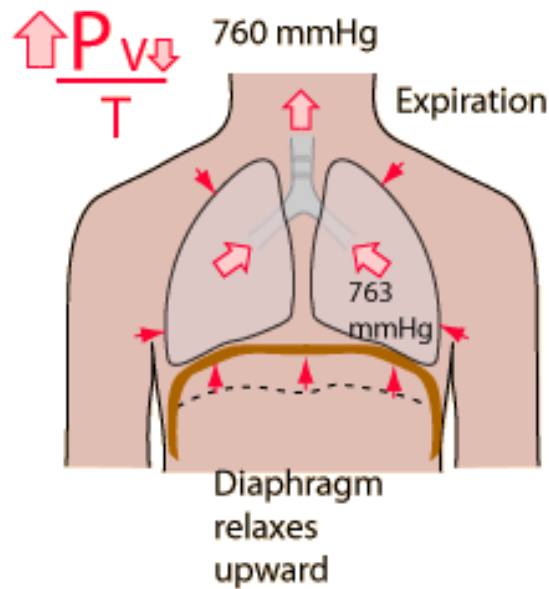
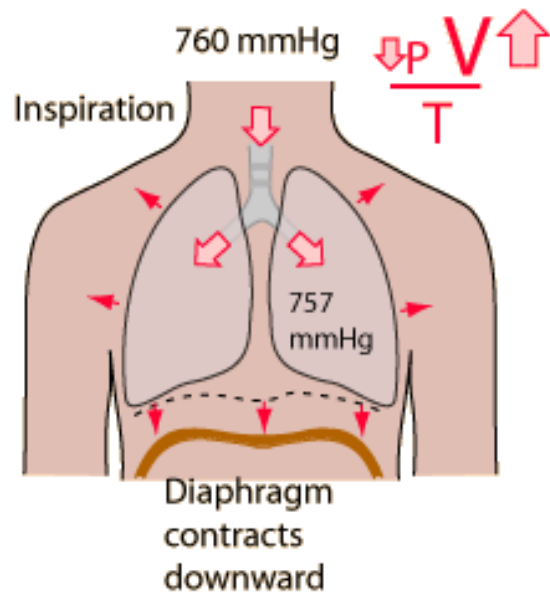
At Rest



A. At rest

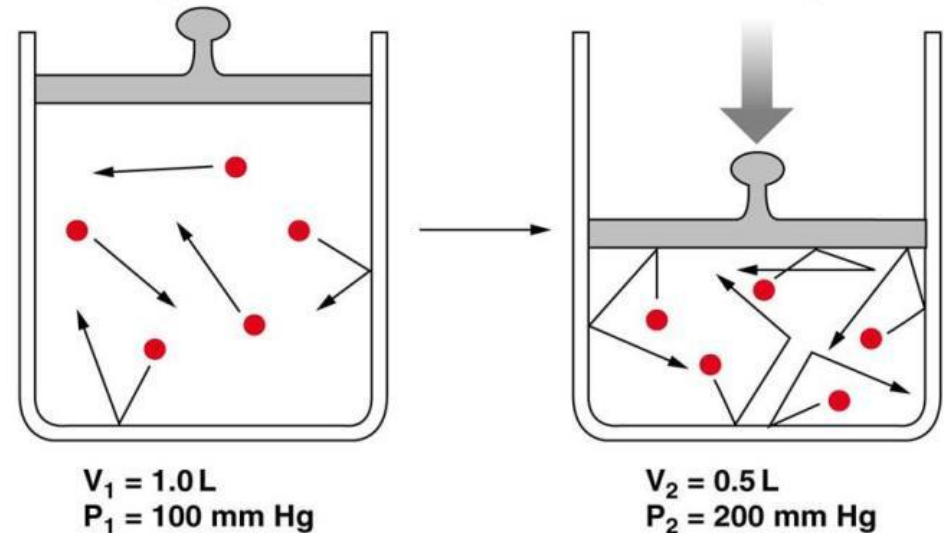
1. Respiratory muscles are at rest.
2. Recoil of lung and chest wall are equal but opposite.
3. Pressure along tracheobronchial tree is atmospheric.
4. There is no airflow.





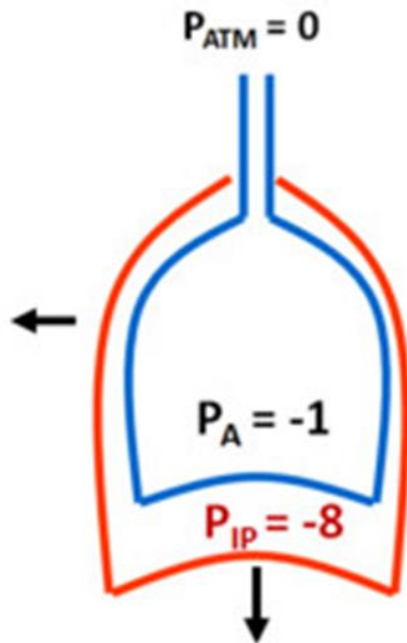
Boyle's Law: $P_1V_1 = P_2V_2$

Decreasing volume increases collisions and increases pressure.



2. Alveolar Pressure

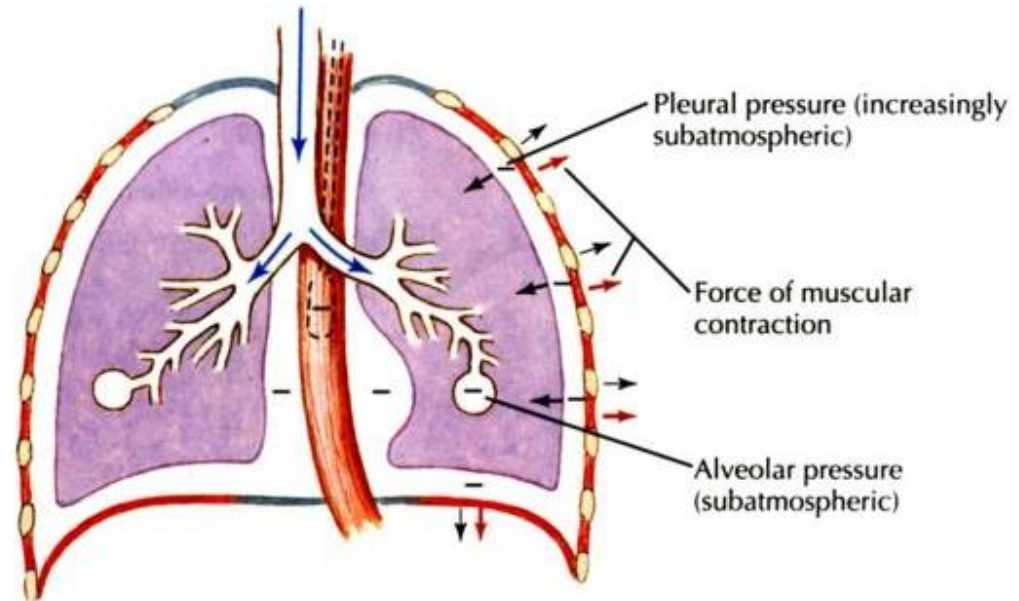
Pressure Changes during Inspiration



1. Inspiratory muscles contract – diaphragm and accessory muscles
2. The thoracic cavity expands due to this muscle contraction
3. The chest wall pulls on the pleural space and generates a more negative P_{IP} (-5 to -8)
4. The P_{IP} pulls on the lungs and expands them
5. The lungs increase in volume and therefore the P_A drops to sub-atmospheric levels (from 0 to -1)
6. Air flows down the $P_{ATM} \rightarrow P_A$ gradient until that gradient is 0 ($P_{ATM} = P_A$; flow stops)

B. During inspiration

Inspiratory muscles contract and chest expands; alveolar pressure becomes subatmospheric with respect to pressure at airway opening. Air flows into lungs.

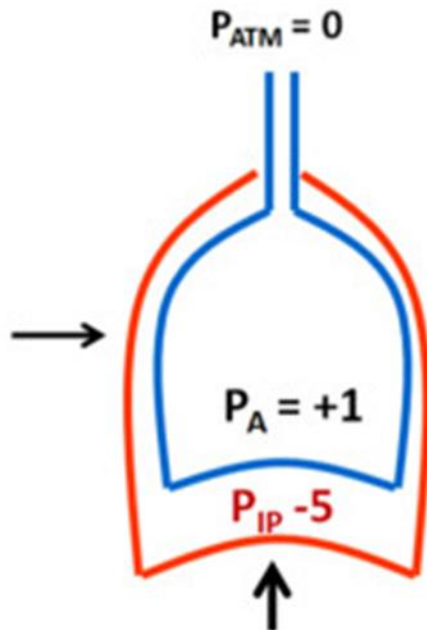


Lung Force Relationships During Inspiration

- Increase in negativity of intrapleural pressure due to contraction of the diaphragm and expansion of the chest wall
- Expansion of chest wall causes gas in the alveoli to expand creating negative alveolar pressure
- Increase in negative pressure in the lung (i.e. alveoli) causes air to flow in to the lungs

2. Alveolar Pressure

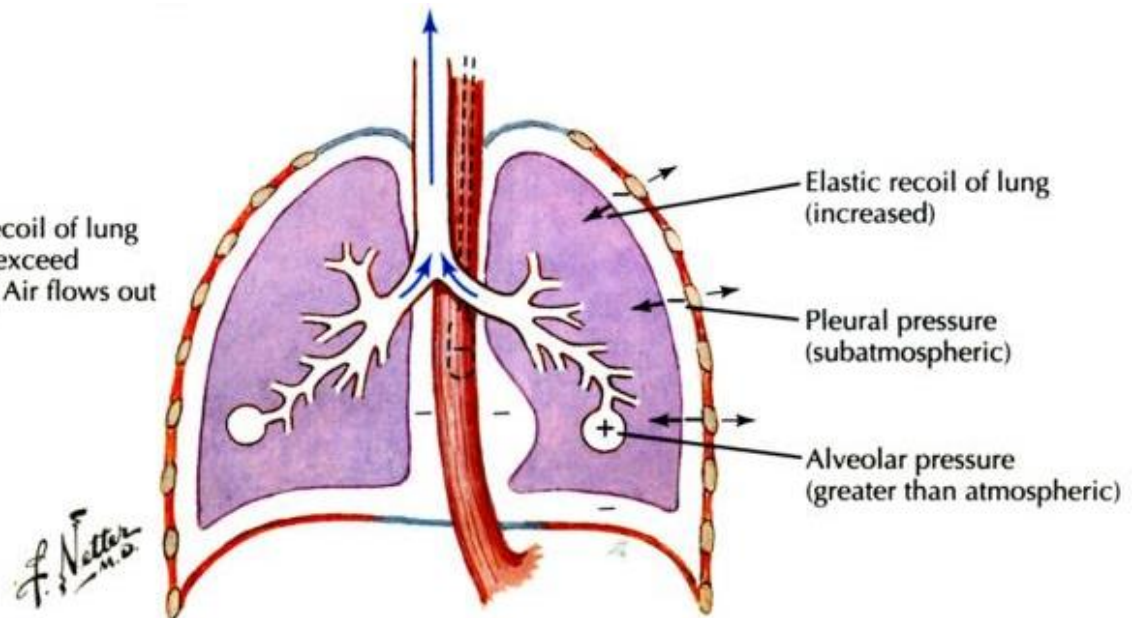
Pressure Changes during Expiration



1. The muscles totally relax
2. The thoracic volume declines
3. The P_{IP} becomes less negative (from -8 to -5 cmH_2O) since the chest wall is no longer pulling as hard on the pleural membrane
4. The lungs will decrease in size since the P_{IP} has become less negative (i.e. less vacuum force pulling on lungs)
5. The P_A will rise above atmospheric as the lungs decrease in size (from 0 to +1)
6. Air flows down the $P_A \rightarrow P_{ATM}$ gradient i.e. from high to low pressure

C. During expiration

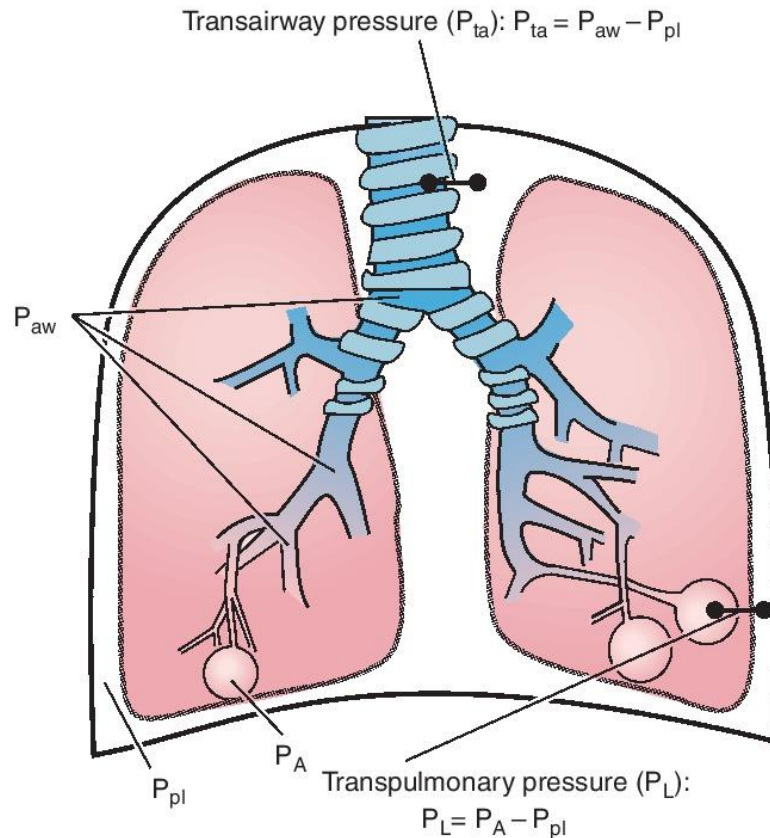
Inspiratory muscles relax; recoil of lung causes alveolar pressure to exceed pressure at airway opening. Air flows out of lung.



Lung Force Relationships During Expiration

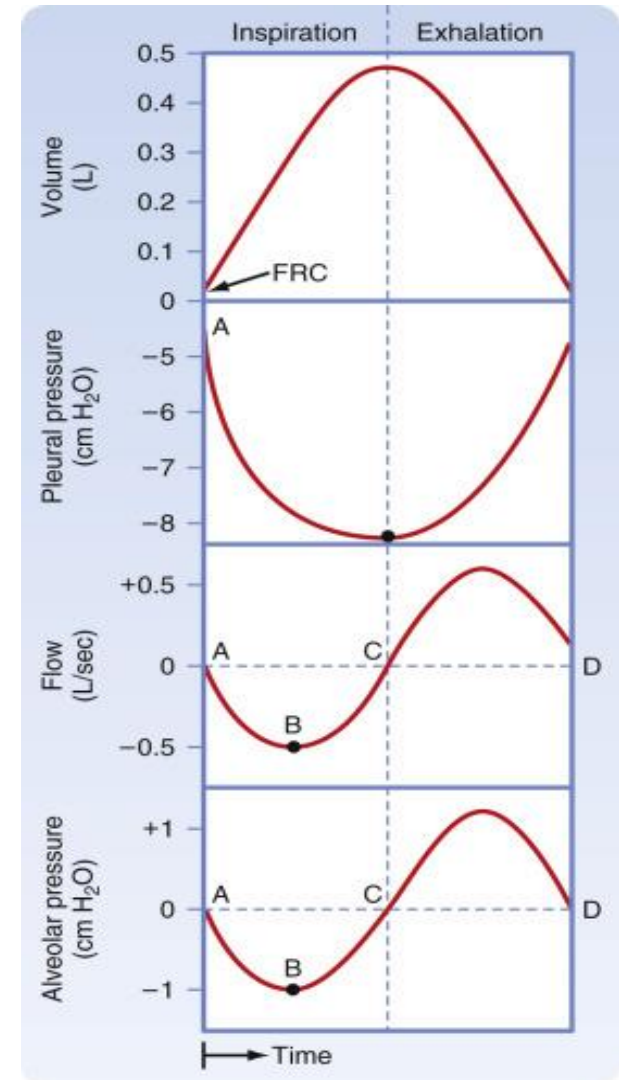
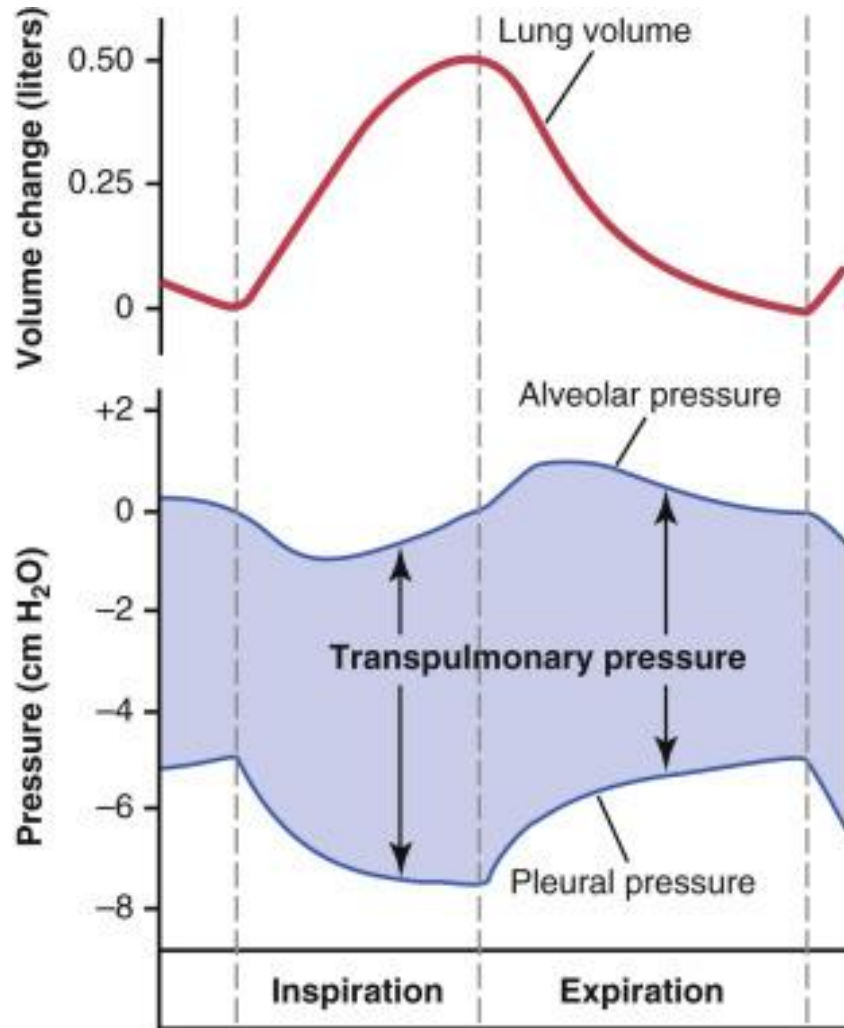
- Relaxation of the muscles of inspiration causes intrapleural pressure to return to -5 cm H₂O under resting conditions.
- Lung deflation begins and continues until the recoil force decreases to again equal intrapleural pressure. Once this occurs, the lung system is back to FRC.
- Deflation of the lung compresses the gases in the alveoli, creating a slightly positive alveolar pressure. This causes air to flow out of the lungs.
- The outflowing air returns alveolar pressure towards zero; at zero, airflow stops.

3. Transpulmonary pressure



- Several important pressures are involved in breathing: airway pressure (P_{aw}), **alveolar pressure (P_A)**, **pleural pressure (P_{pl})**, **transpulmonary pressure (P_L)**, and transairway pressure (P_{ta}). Both **transpulmonary pressure** and transairway pressure can be defined as the pressure inside minus the pressure outside. In both cases, the pressure outside is pleural pressure (P_{pl})
- **P_L** - Measure of the elastic forces in the lungs that tend to collapse the lungs at each instant of respiration, called the ***elastic recoil pressure***

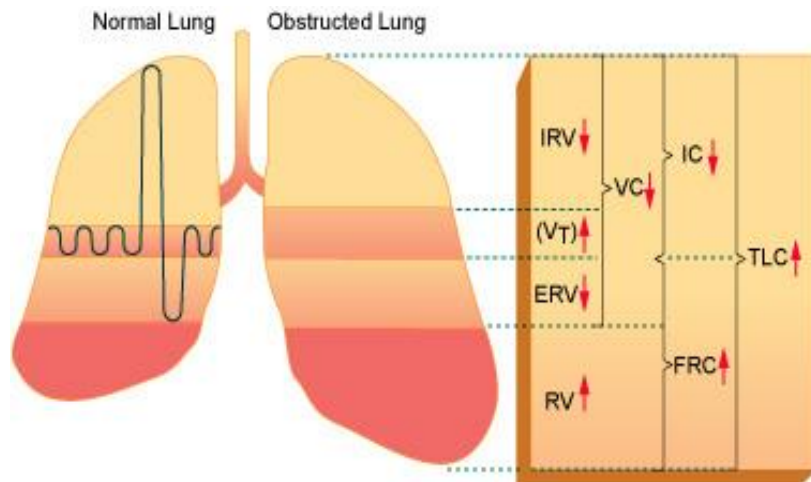
Pressures Acting on the System



Summary

- The outward elastic recoil of the chest wall (pleural pressure – pressure at surface of chest) is opposed by the inward elastic recoil of the lungs (alveolar pressure – intrapleural pressure).
- **Transpulmonary pressure** is (**elastic recoil pressure**) the **difference** between the **alveolar pressure** and the **intrapleural pressure** in the lungs. During ventilation, air flows because of pressure gradients.

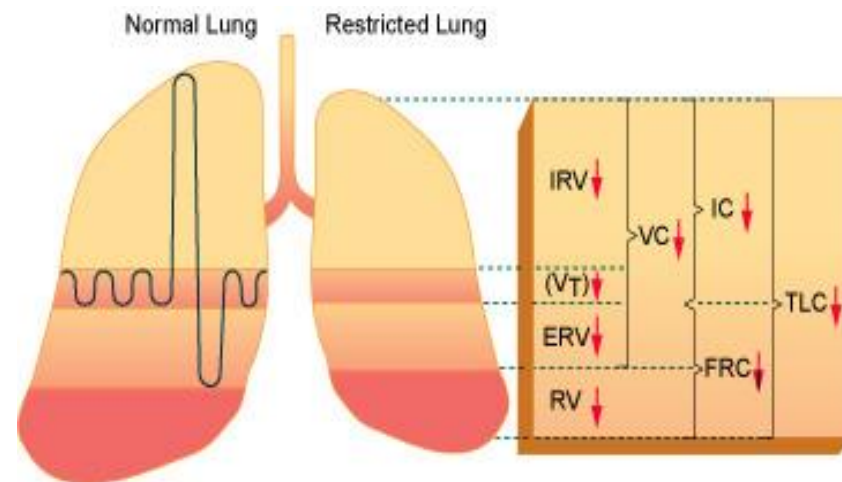
Obstructive Lung Disease



***Loss of ability to develop
+ alveolar pressure***

- Increased RV and FRC (both of these represent air left in the lung after expiration).

Restrictive Lung Disease



***Loss of ability to develop
- alveolar pressure***

- Decreased FRC (because less air is taken in with inspiration)
- Decreased TLC (because lungs are hard to expand and can't "stretch" as much on inspiration)

Quick Quiz

As inspiration proceeds, the pleural pressure reaches until:

- A. -4 cm H₂O
- B. +2 cm H₂O
- C. 0 cm H₂O
- D. -8 cm H₂O
- E. +6 cm H₂O

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4. Chart the functional relationship between intrapleural, alveolar and transpulmonary pressures.
5. Interpret relevant lung biomechanics in relation to restrictive and obstructive lung diseases.

More sample questions for practicing

1. <https://exchange.scholarrx.com/brick/physiology-of-ventilation-and-respiration>
2. <https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=258031997&bookid=3310>
3. <https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=49980261&bookid=811>
4. <https://casefiles-mhmedical-com.nyit.idm.oclc.org/CaseContent.aspx?gbosID=218980&gbosContainerID=68&viewByNumber=false&groupid=0#104658253>

References

- **Guyton and Hall's Textbook of Medical Physiology**, 14th and 15th Eds.; Chapters 38, 40, 43 (as applicable to lecture extent)
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 - <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20170004883>
 - <https://www.clinicalkey-com.nyit.idm.oclc.org/#!/browse/book/3-s2.0-C20220008424>
- **Boron and Boulpaep's Medical Physiology**, 3rd Ed. 2017; Chapters 27 (as applicable to lecture extent)
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 - <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20110061677>
- **Nunn's applied respiratory physiology**, 9th Ed. 2020; Chapter 2, 5 (as applicable to lecture extent)
 - Copy & paste should work; direct-click may not work sometimes
 - <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20180006702>
- **ScholarRx Brick(s) and link(s):**
 - Physiology of Ventilation and Respiration
 - <https://exchange.scholarrx.com/brick/physiology-of-ventilation-and-respiration>



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